

Q-Learning

Kianté Brantley
CS 1840 Fall 2026

Announcements

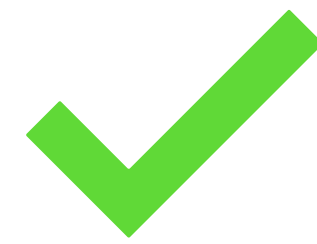
HW2 is out and is due 10/8 (this Thursday)!

HW3 will be released on Friday!

Check the course website for Office Hours:

<https://harvard-cs-1840.github.io/2026/>

Finishing Up From Last Lecture



1. Simple Monte Carlo methods

2. Temporal Difference Learning

Recap: MC estimation with incremental update

Instead of waiting for all N traj, we can also update the estimator incrementally every traj

Initialize $\hat{V}^{\pi}(s) = 0, \forall s$

For all state $s \in \mathcal{S}$:

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η : learning rate

total reward of one traj

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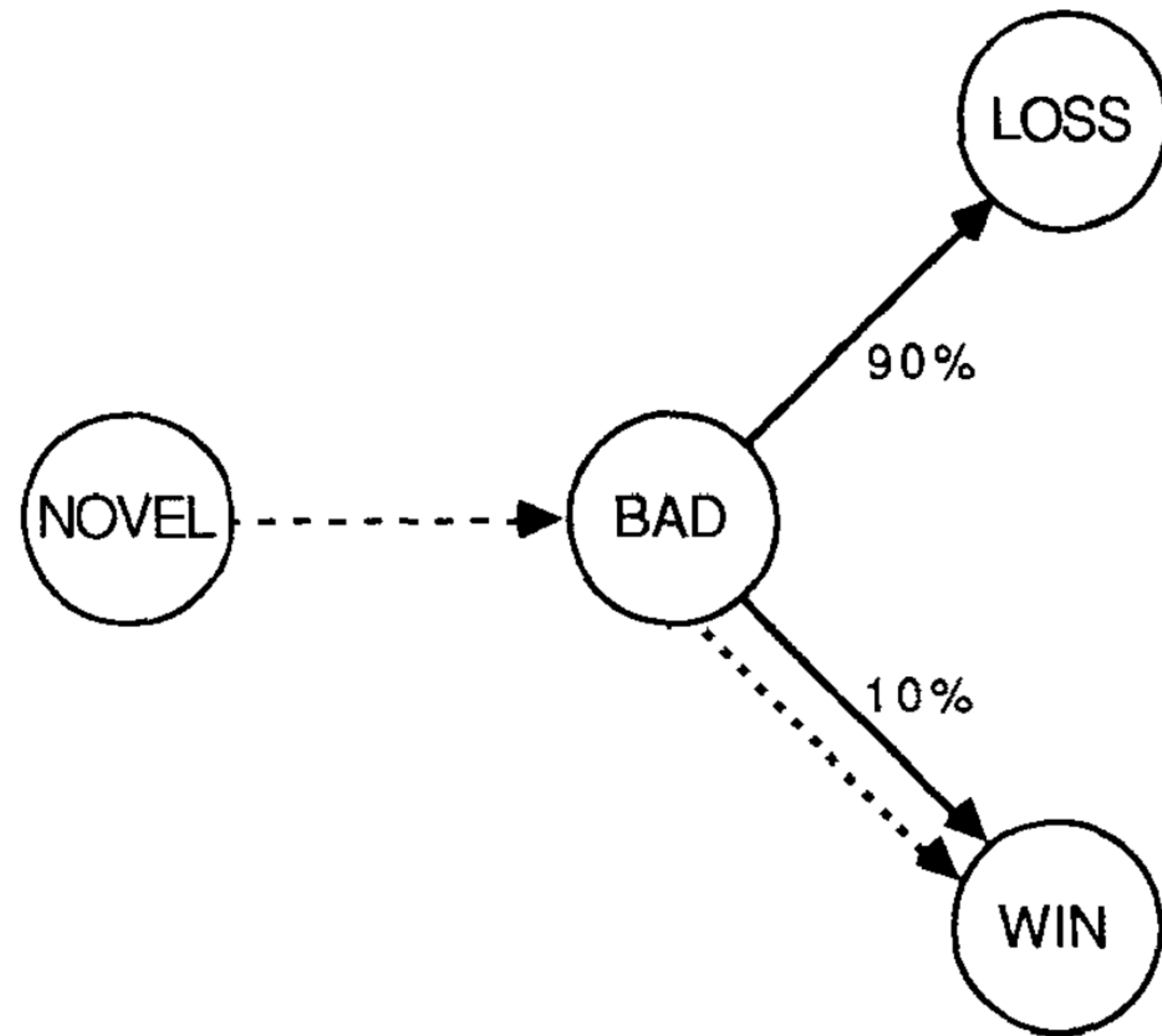
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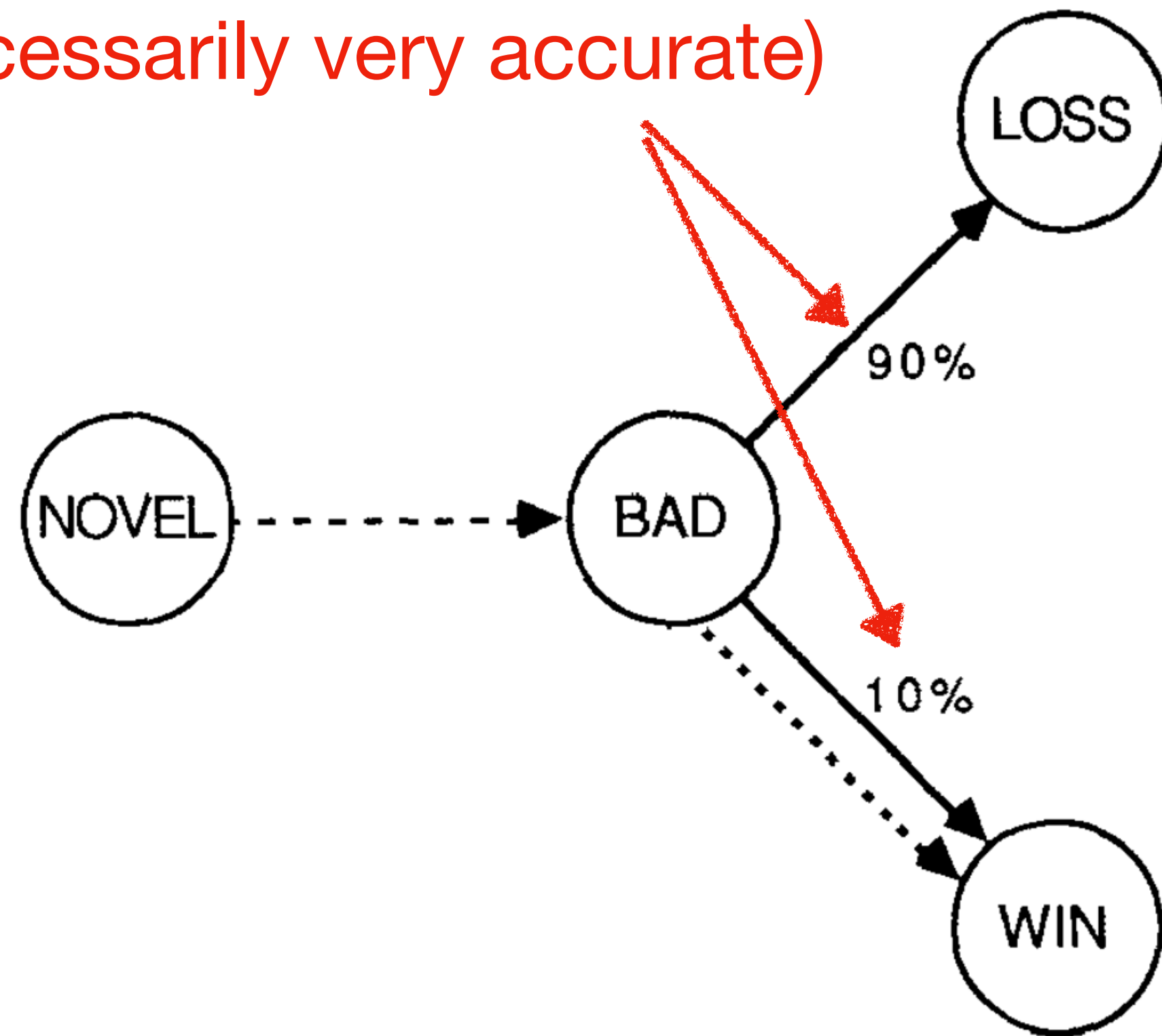
Example of faster learning with TD



[Sutton, 1988]

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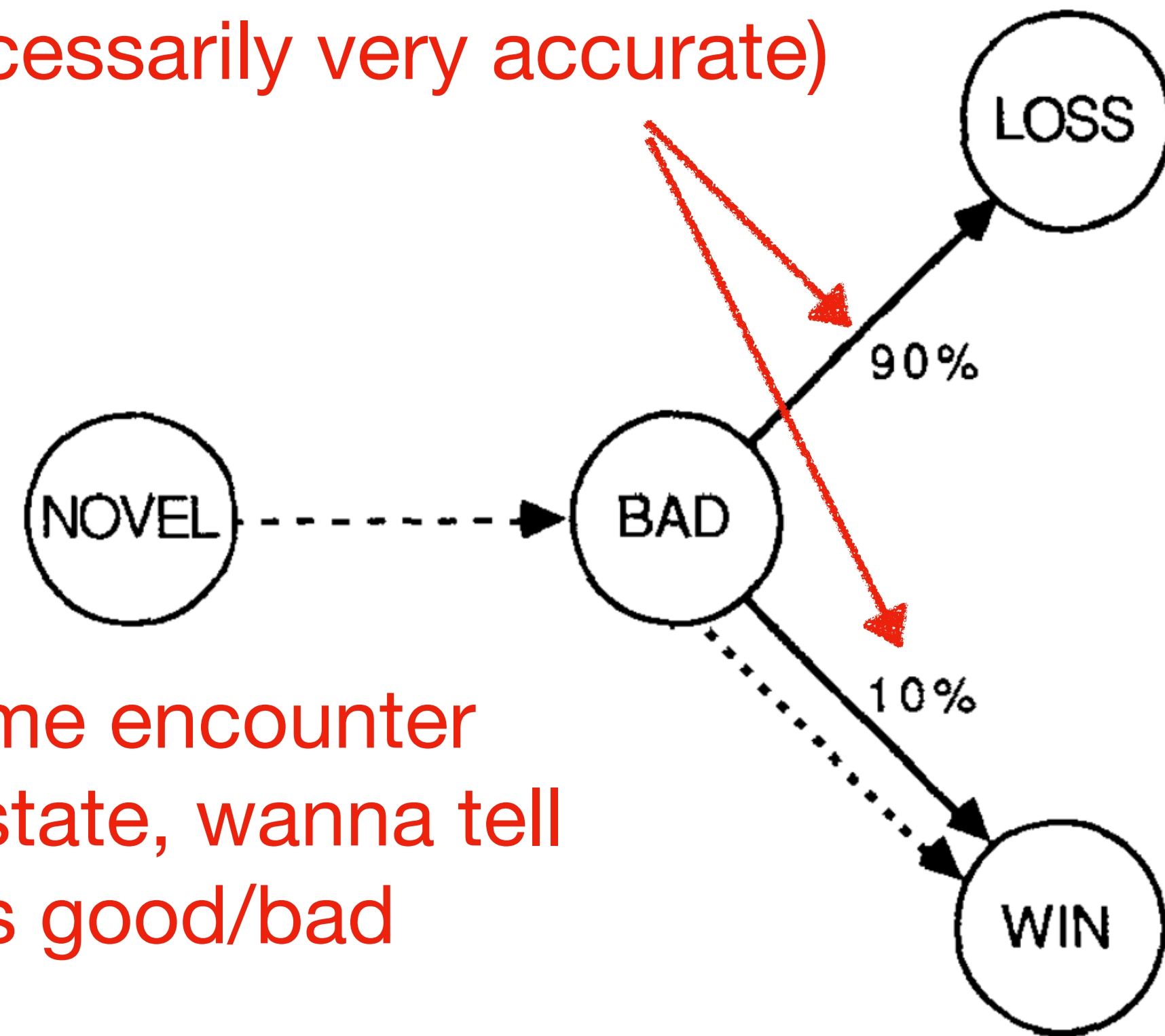
From past game experiences
(not necessarily very accurate)



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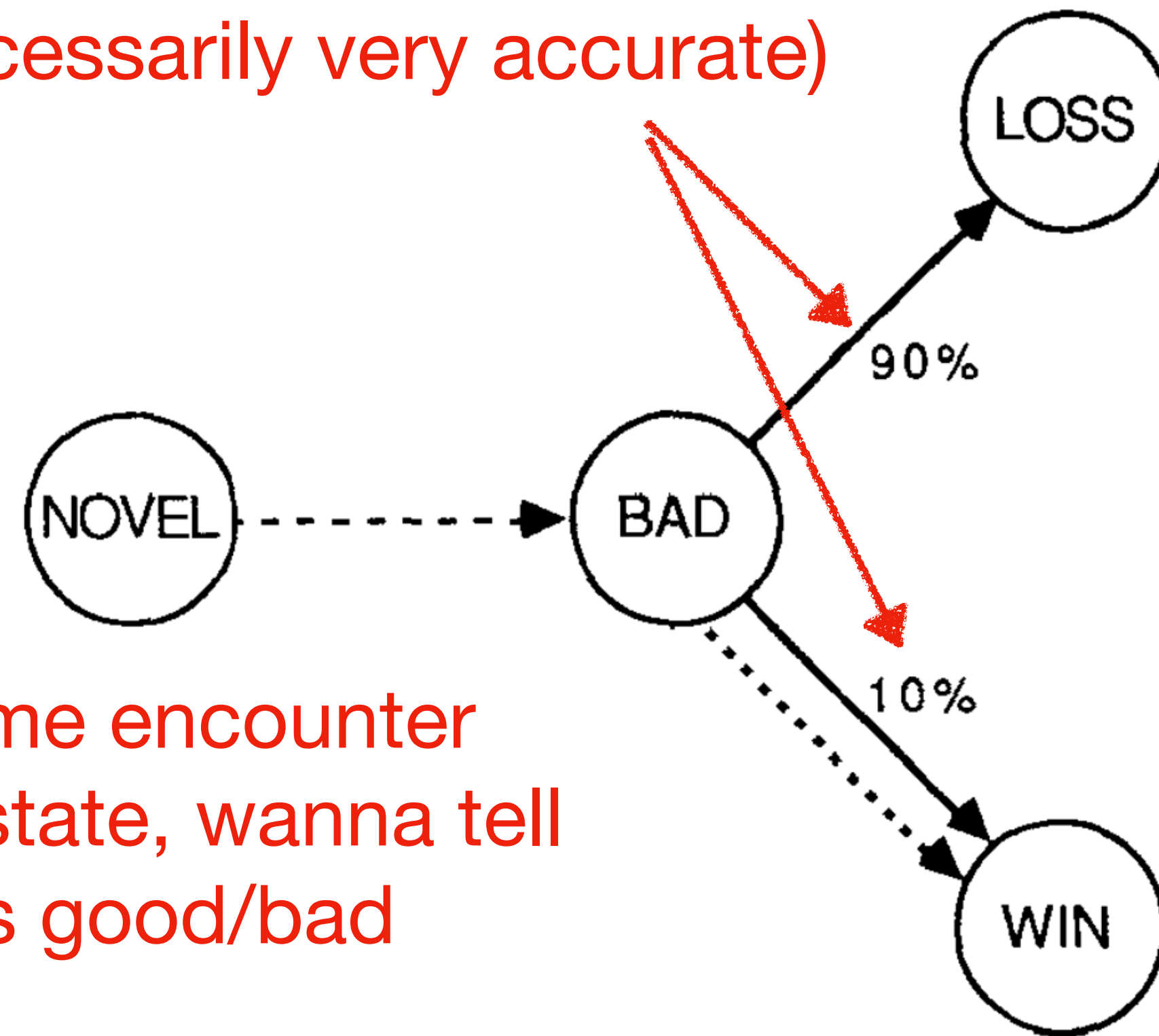


first time encounter
“novel” state, wanna tell
if it’s good/bad

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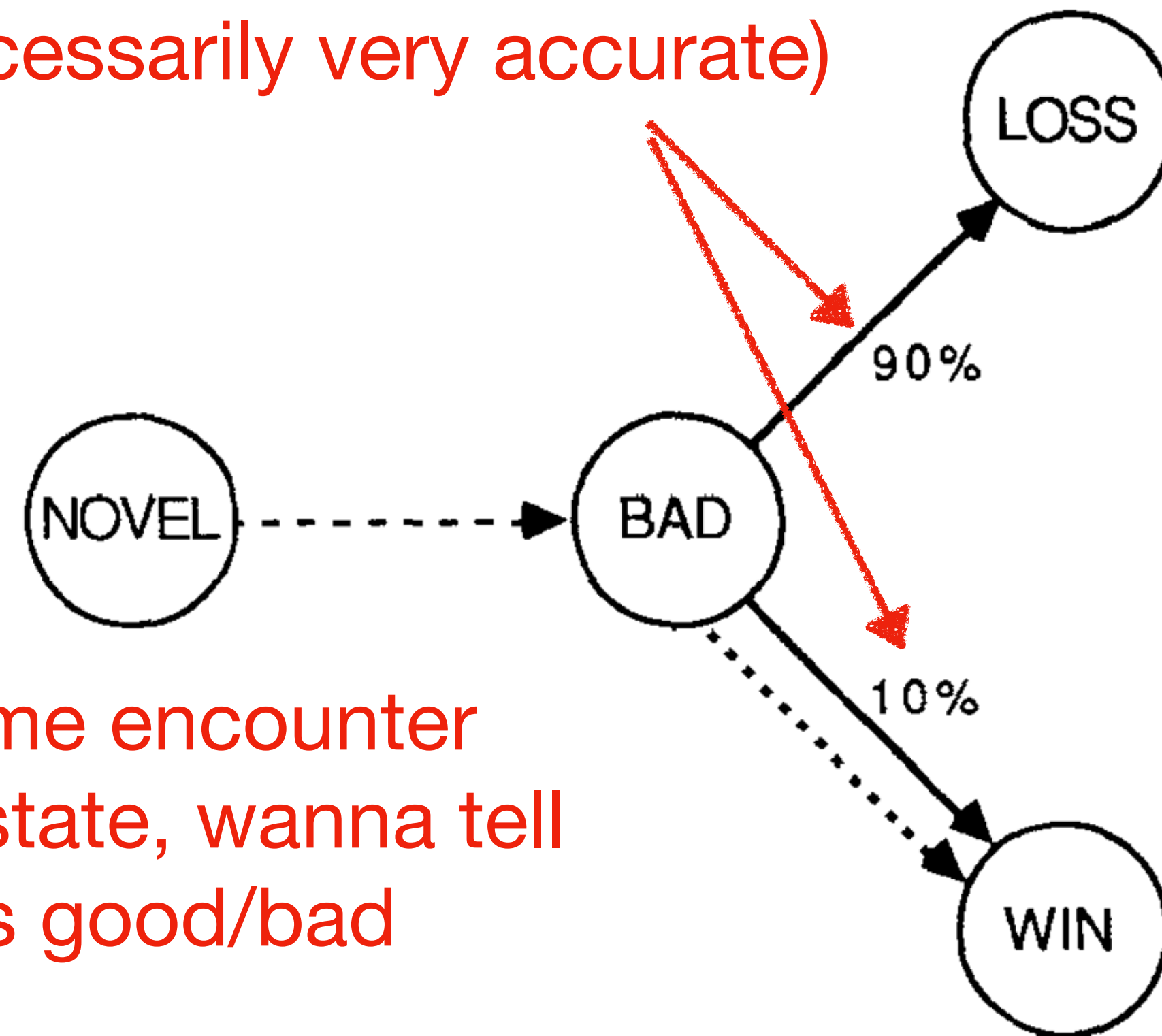
MC: have to rollout from NOVEL many times
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MC: have to rollout from NOVEL many times
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TD (bootstrapping): NOVEL leading to BAD
in one-step implies NOVEL is likely bad

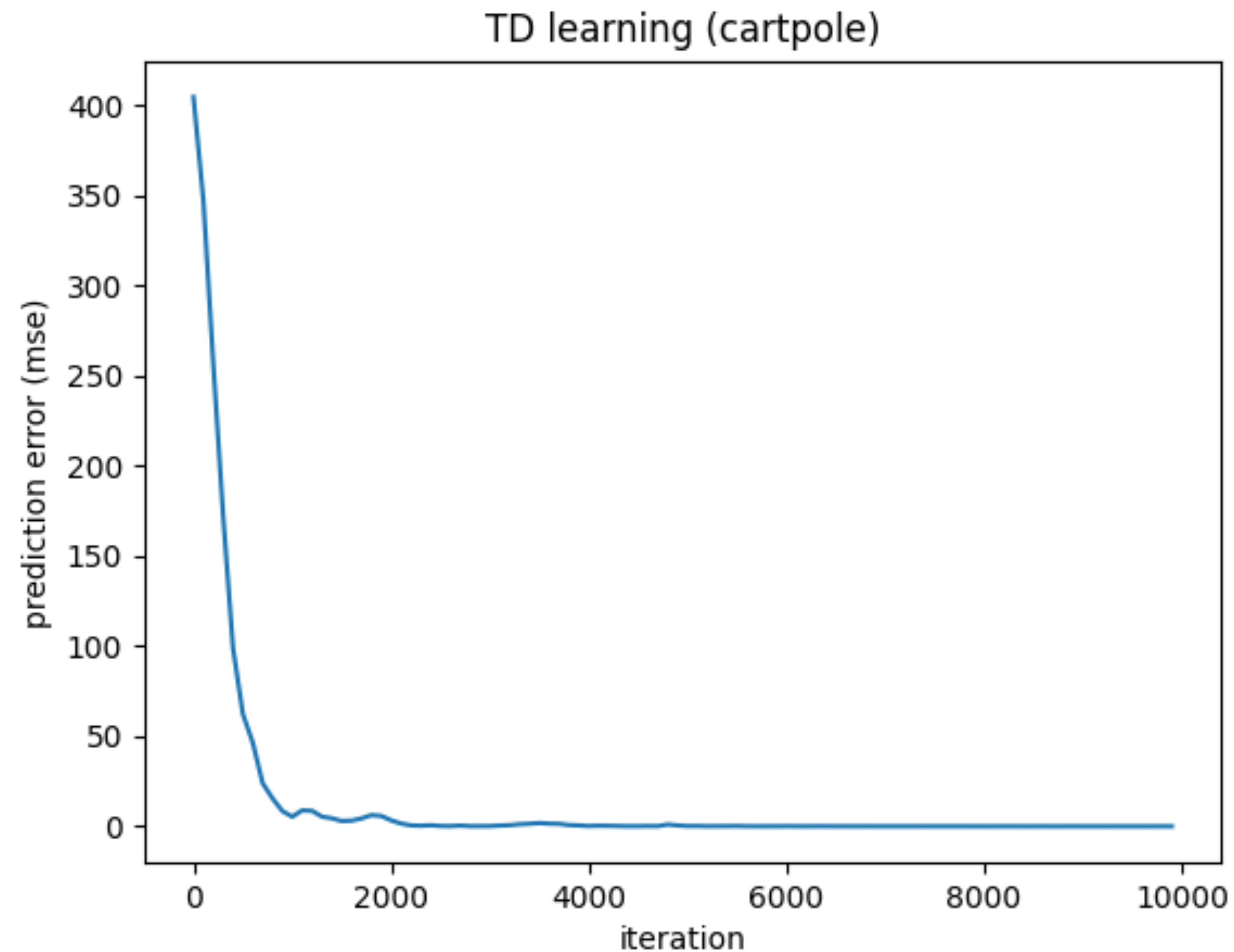
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TD learning on CartPole

Note: Cartpole's state is continuous, so we will need TD w/ function approximation, e.g., neural network (we will get there very soon)

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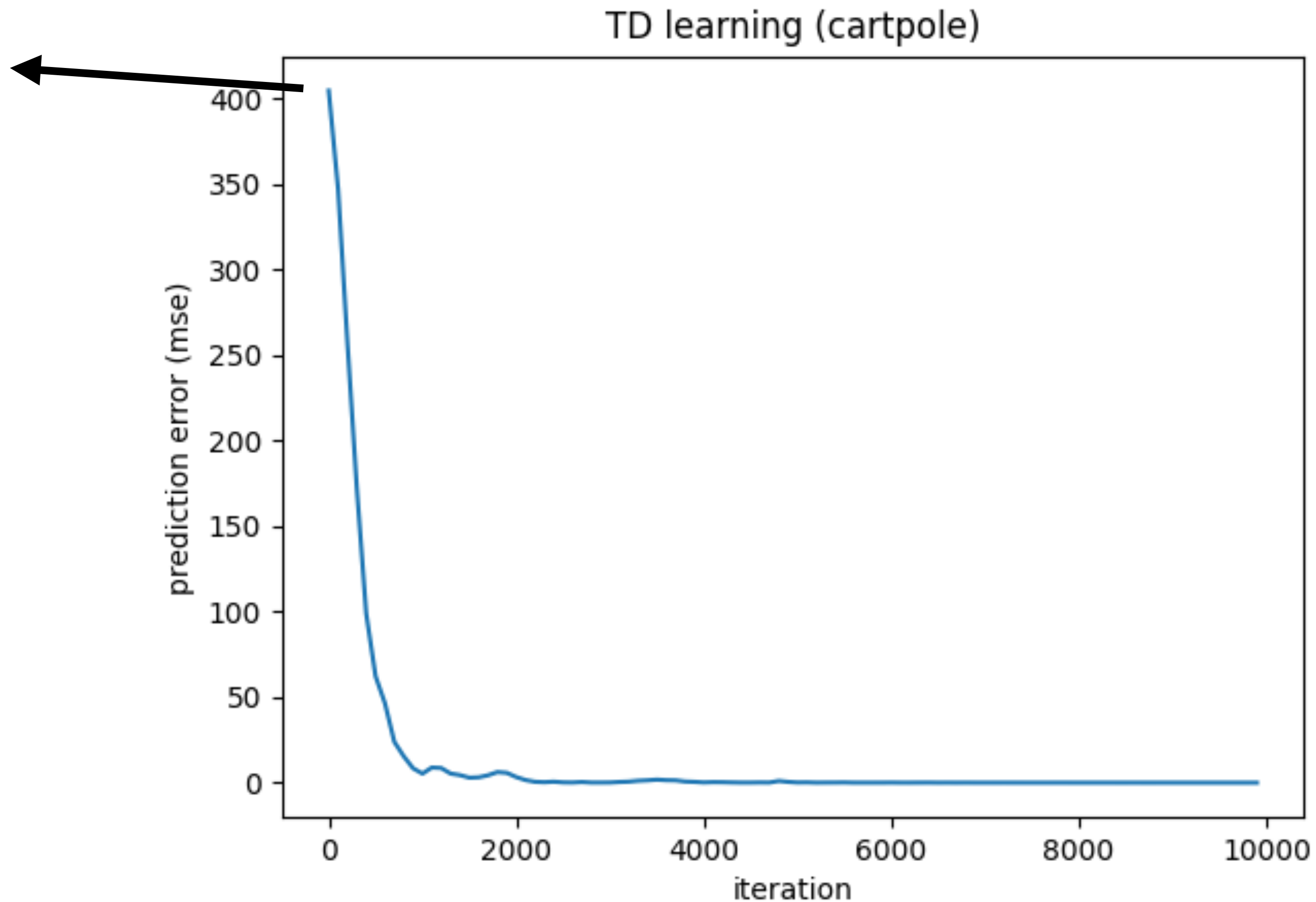
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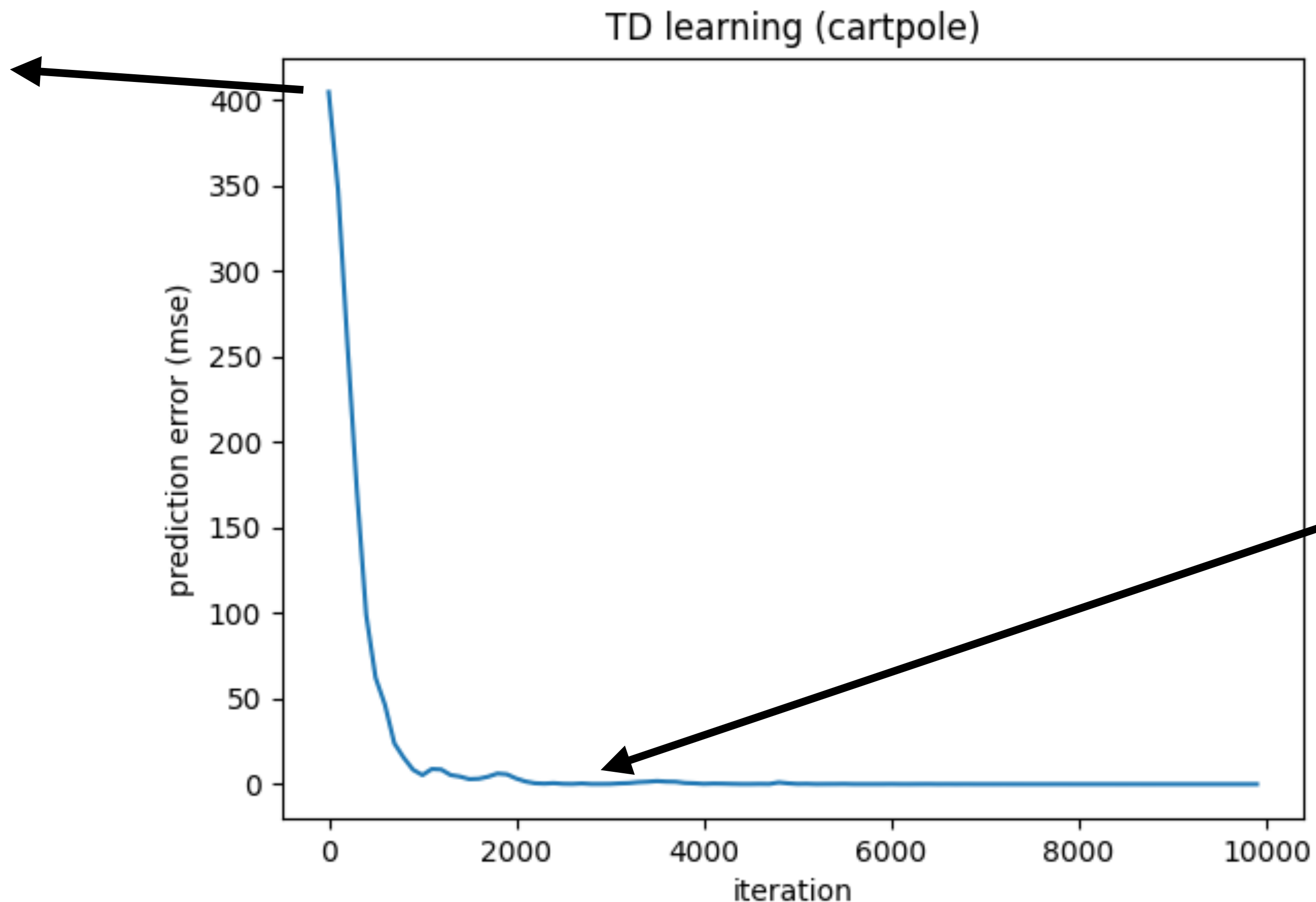
A randomly initialized NN as $\hat{V}^\pi(s)$ (very bad estimator)



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A randomly initialized NN as $\hat{V}^\pi(s)$ (very bad estimator)



Almost perfectly estimate the true target $V^\pi(s)$

TD vs MC: bias-variance tradeoff

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Randomness in target is
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TD vs MC: bias-variance tradeoff

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Large bias when $\hat{V}^\pi(s')$ is far from $V^\pi(s')$

$$\hat{V}^\pi(s) \Leftarrow \hat{V}^\pi(s) + \eta \left(\underline{r + \gamma \hat{V}^\pi(s')} - \hat{V}^\pi(s) \right)$$

Randomness in target is from **one-step transition**

Recap: Questions

Can we extend the one-step TD target to include multiple future rewards before bootstrapping?

TD relies on the idea of **Bootstrapping** — *using predictions from our estimator to update the estimator itself*

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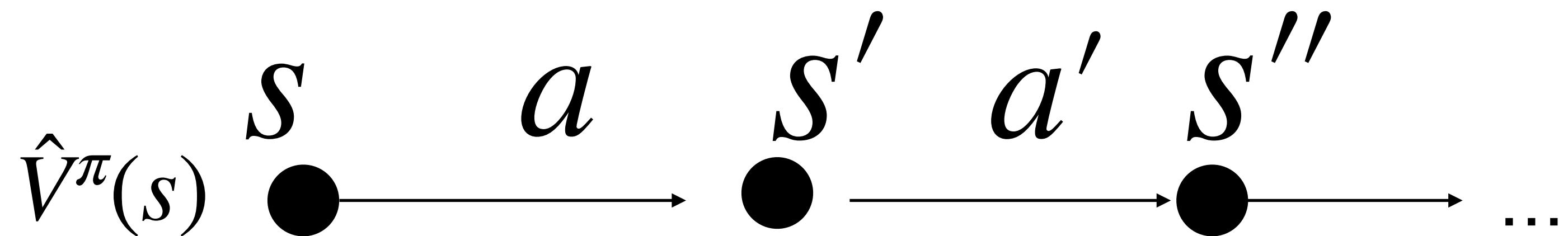
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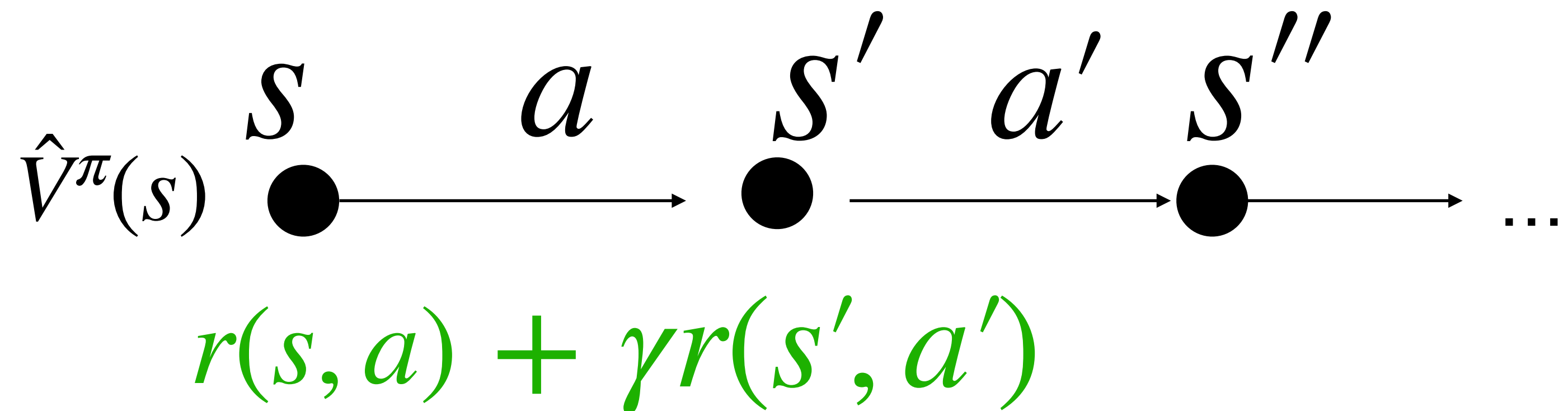
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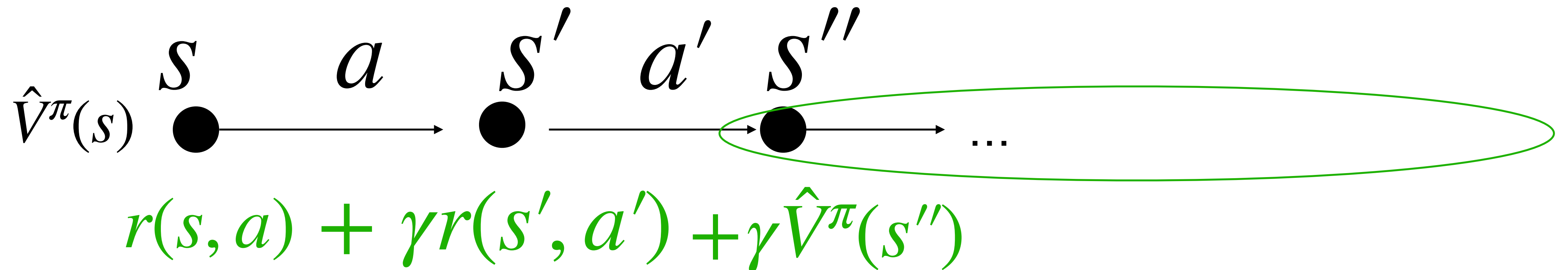
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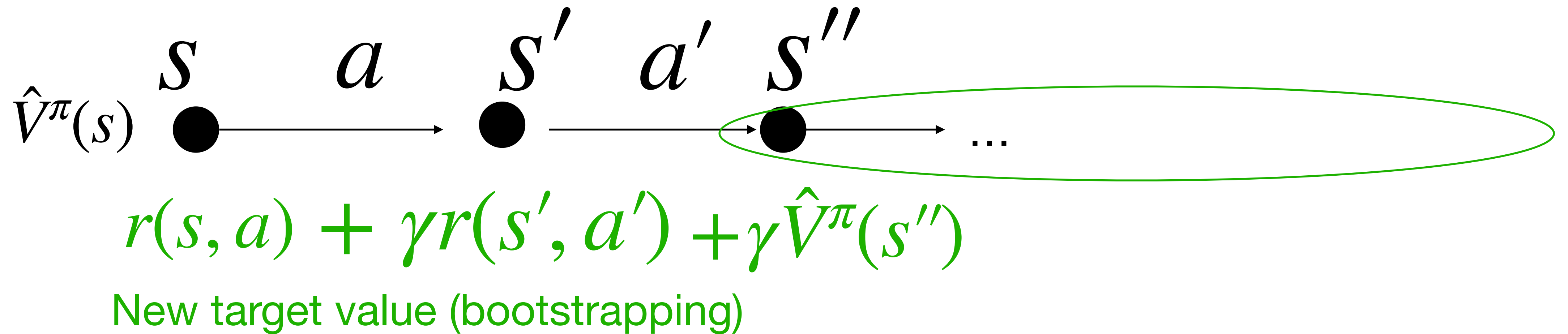
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Recap: Bellman Optimality

Bellman Optimality

$$Q^{\star}(s, a) = r(s, a) + \gamma \mathbb{E}_{s' \sim P(\cdot | s, a)} \max_{a'} Q^{\star}(s', a'), \forall s, a$$

VI: An iterative approach for estimating $Q^{\star}$

$$Q \leftarrow \mathcal{T} Q$$

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1. Need to know the transition
2. Only works for discrete small MDPs

Recap: Bellman Optimality

Q: if there is some $Q(s, a)$, such that the following holds:

$$Q(s, a) = r(s, a) + \gamma \mathbb{E}_{s' \sim P(\cdot | s, a)} \max_{a'} Q(s', a'), \forall s, a$$

is this $Q = Q^*$?

Today

Given MDP $\mathcal{M} = (S, A, r, P, \gamma)$,

**how to estimate $Q^*(s, a), \forall s$ WITHOUT knowing P
i.e., how to learn Q^* (thus, π^*) from experience**

Motivation

Computing a near-optimal policy to achieve the long-term goals w/o knowing or explicitly modeling the world

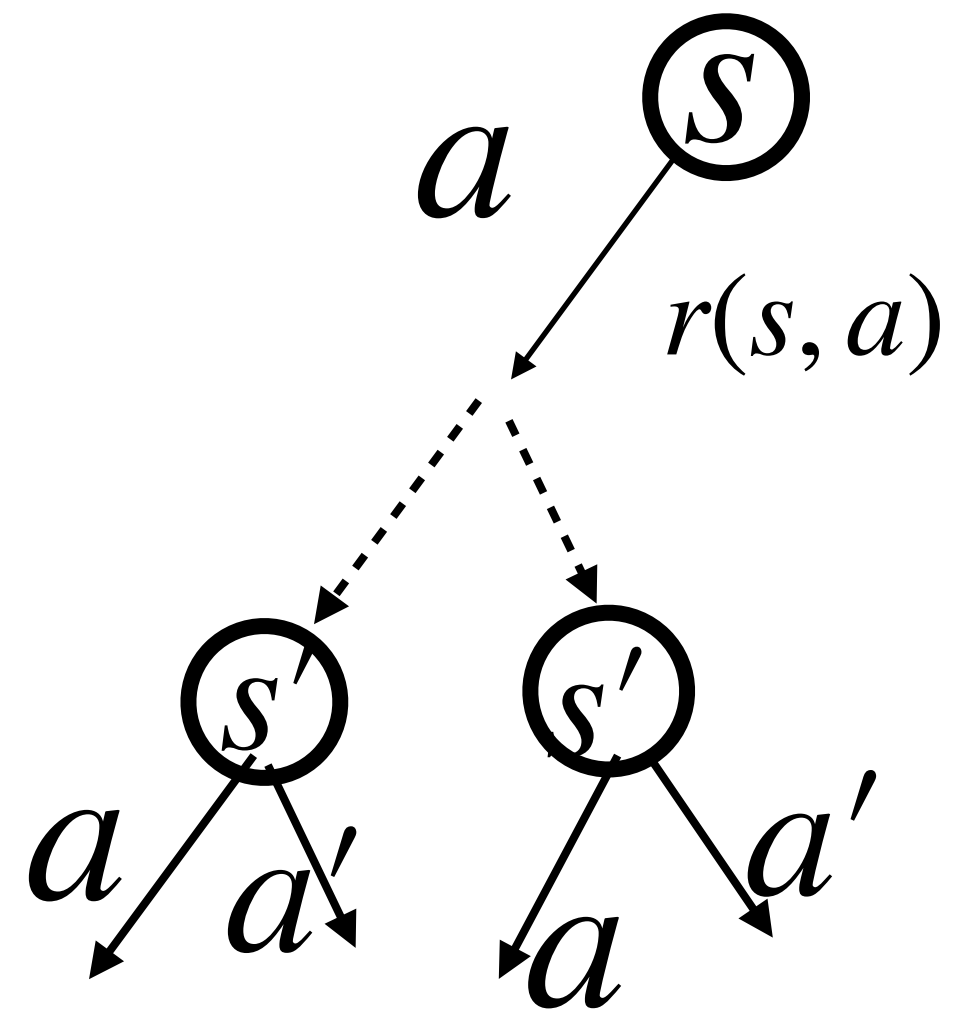


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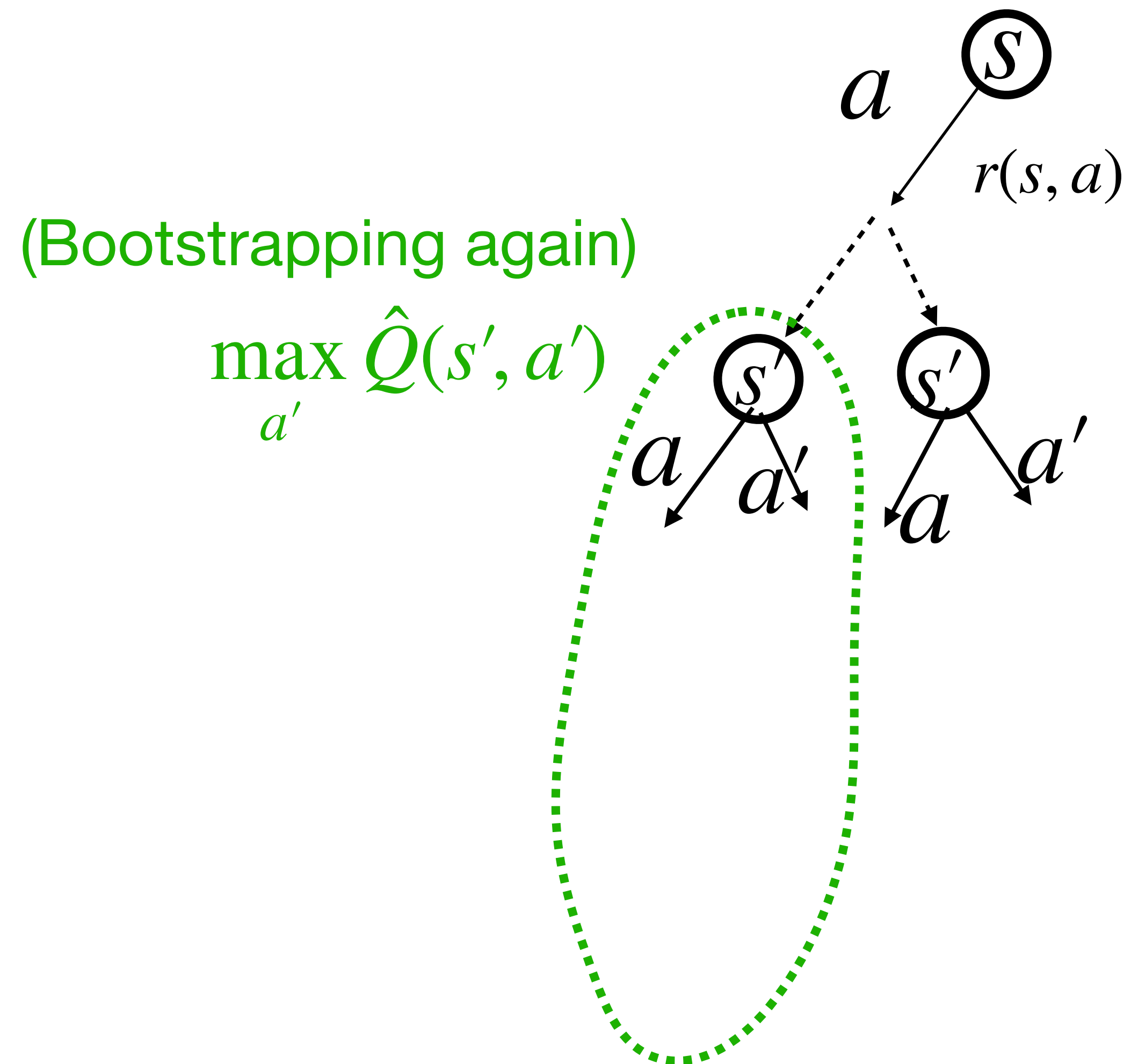
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2. Revisit TD: Off-policy TD Learning

Q Learning



Q Learning



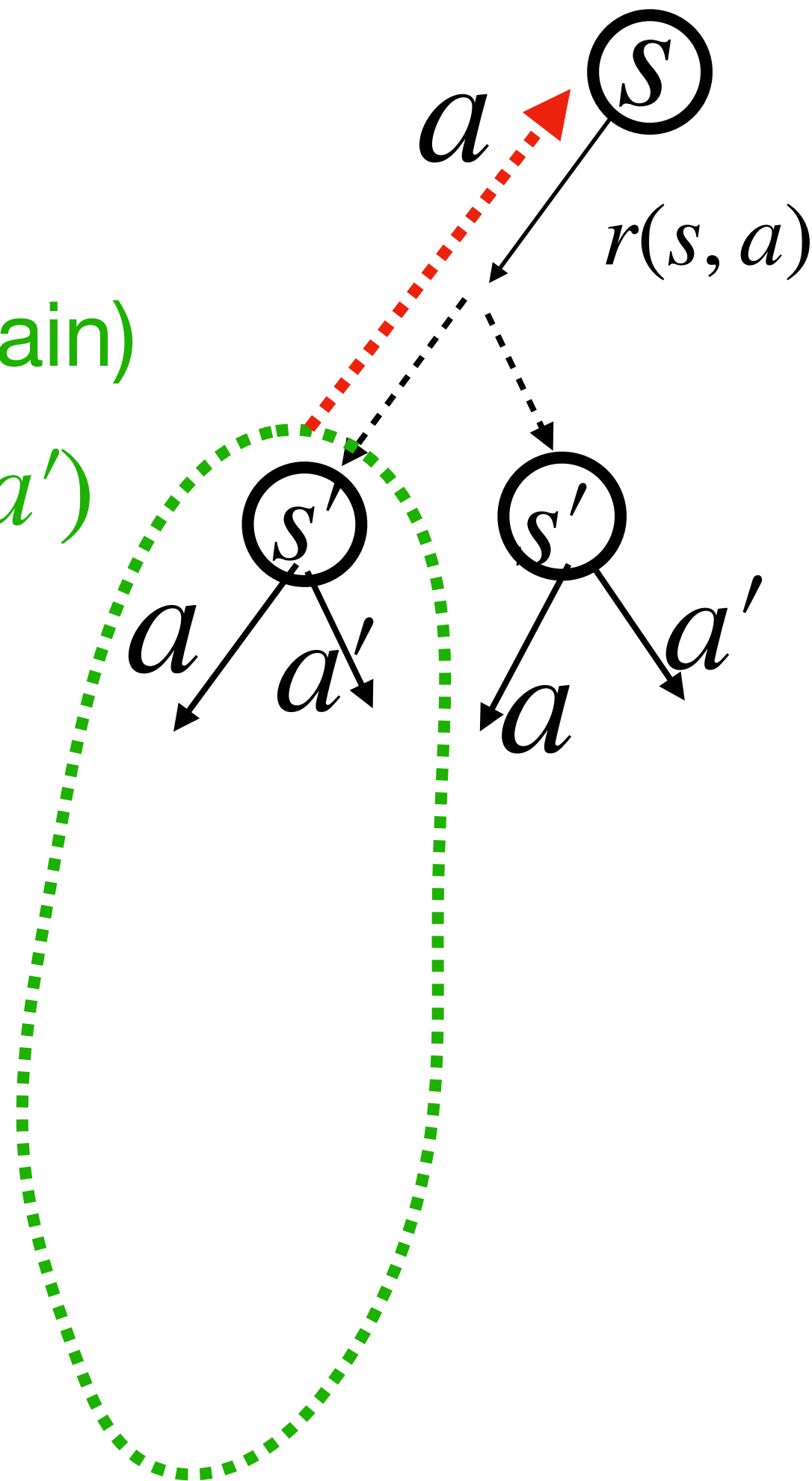
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target

(Bootstrapping again)

$$\max_{a'} \hat{Q}(s', a')$$



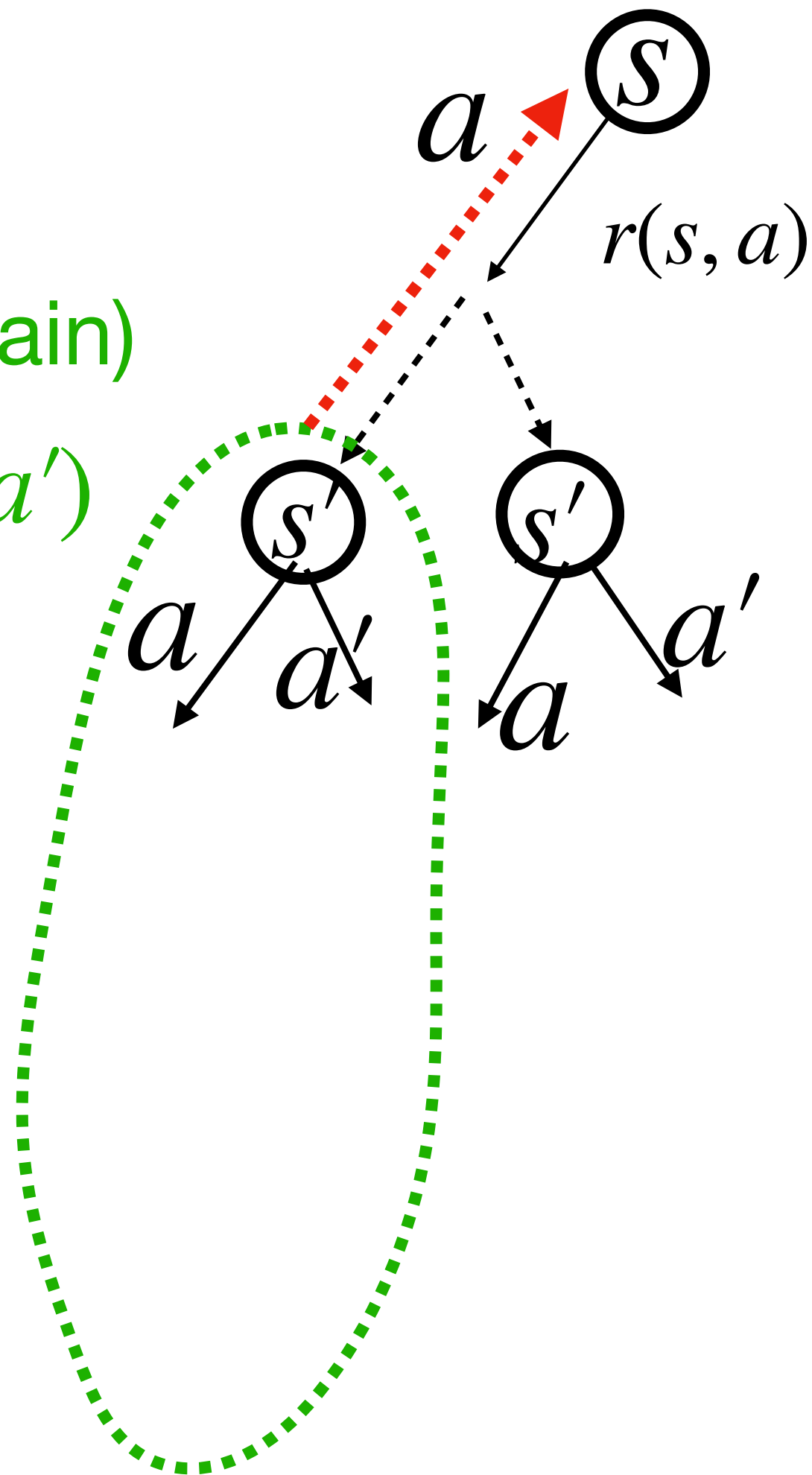
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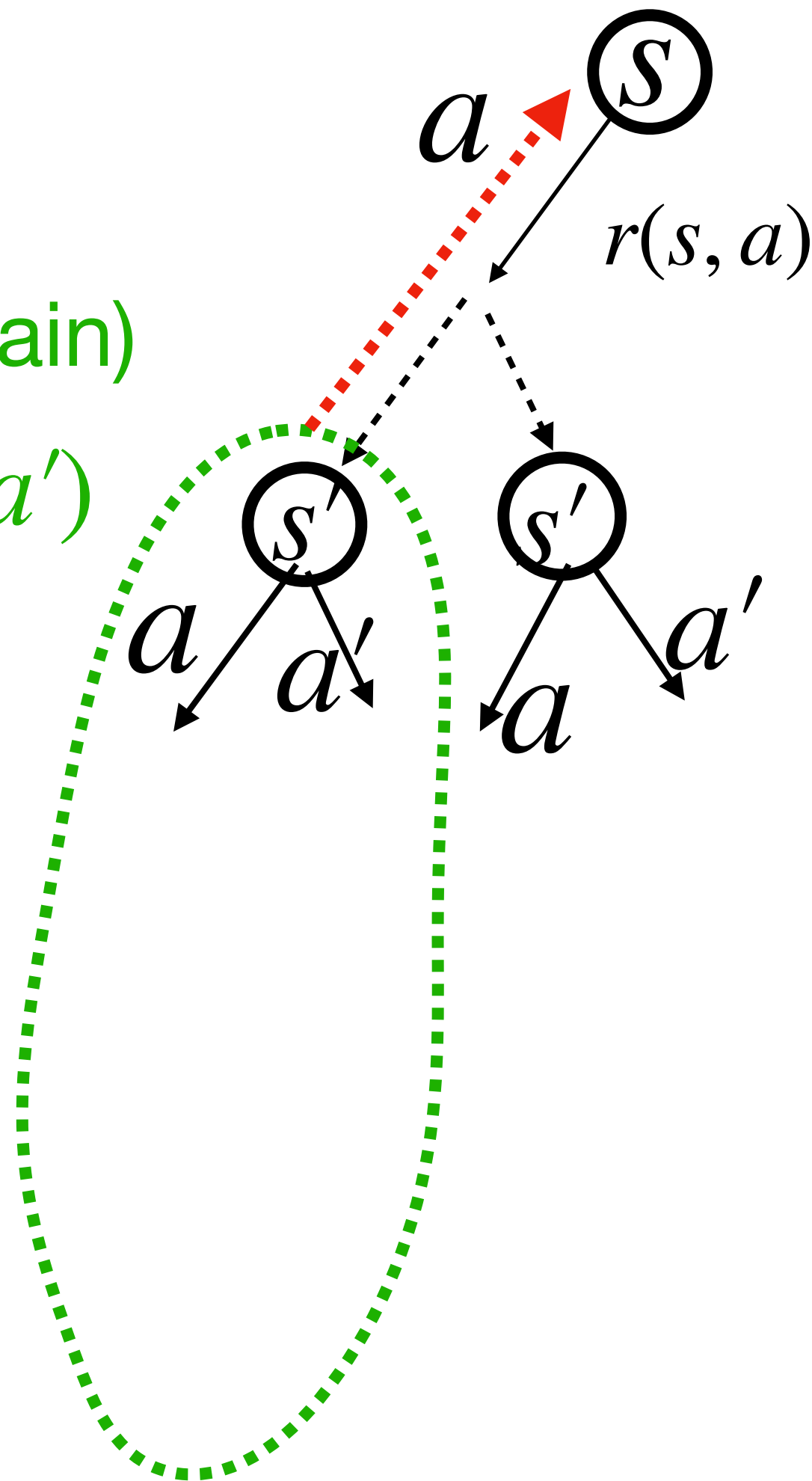
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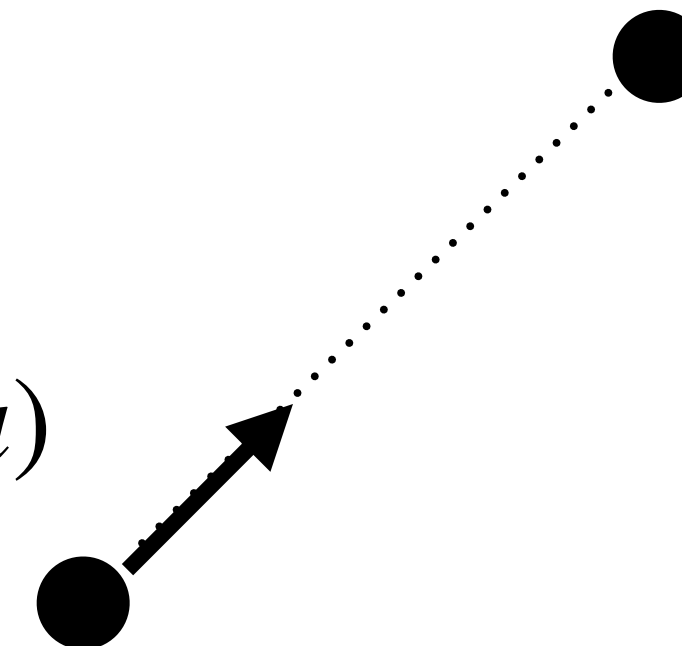
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$$\hat{Q}(s, a)$$

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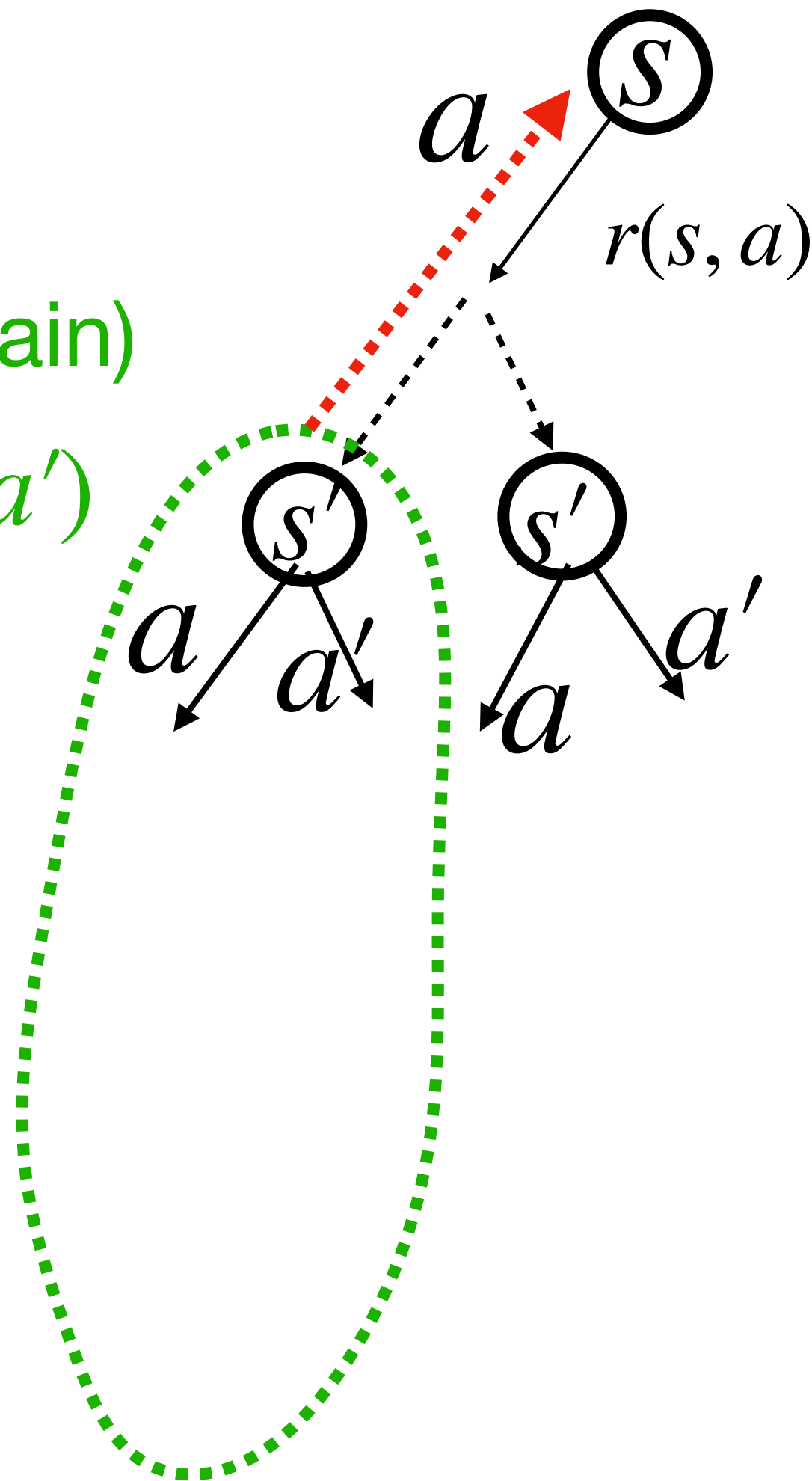


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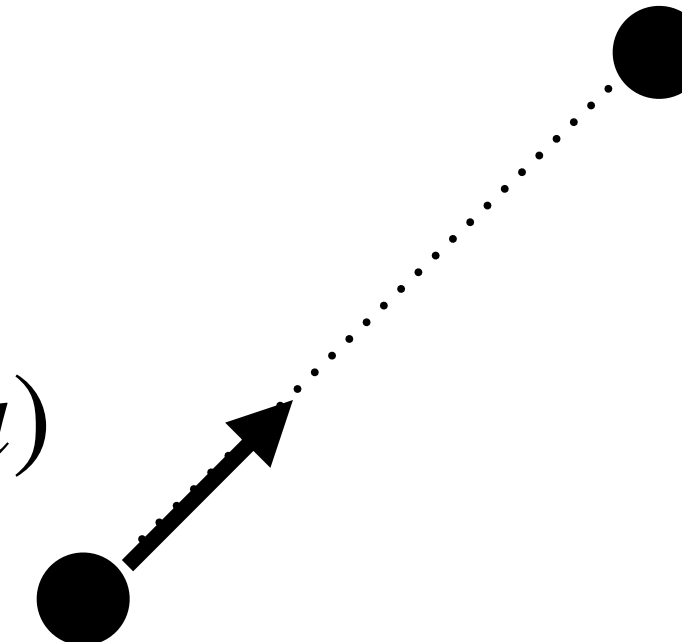
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target

$$\hat{Q}(s, a)$$

Target



Q-learning update: move to the target with a small step

Q Learning

Given a **one-step transition** (s, a, r, s') where $r = r(s, a)$, $s' \sim P(\cdot | s, a)$:

Q-learning updates the guess at (s, a) as follows:

$$\hat{Q}(s, a) \leftarrow \hat{Q}(s, a) + \eta \left(\underbrace{r + \gamma \max_{a'} \hat{Q}(s', a')}_{\text{Q-Target}} - \hat{Q}(s, a) \right)$$

Q-Target
(Constructed via Bootstrapping!)

Q Learning

How to collect data?

Choice one: trust current estimator $\hat{Q}$, always use $\arg \max_a \hat{Q}(s, a)$

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W/ prob ϵ , select action uniform randomly

W/ prob $1 - \epsilon$, select greedy action $\arg \max_a \hat{Q}(s, a)$

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Interpret Q-learning as “SGD” on Bellman error

Q-learning is not the usual SGD, i.e., it is **not** running SGD to minimize a fixed loss function

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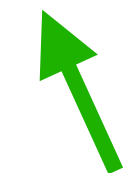
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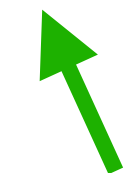
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Unbiased
estimate of y

$$\widetilde{\nabla} \ell_{be}(x) |_{x=\hat{Q}(s, a)} := 2 \left(\hat{Q}(s, a) - \left(r + \gamma \max_{a'} \hat{Q}(s', a') \right) \right)$$

This keeps changing as
we learning

Q Learning Theory

[Informal] Assume the ϵ -greedy strategy has non-trivial probability of visiting every state-action pair. Setting learning rate η properly, we will have:

$$\hat{Q}(s, a) \rightarrow Q^*(s, a), \forall s, a$$

when # of interactions approaches to ∞

(concrete convergence rates are known as well)

Outline:



1. Q Learning

2. Revisit TD: Off-policy TD Learning

TD Learning

Given (s, a, r, s') , where $a \sim \pi(\cdot | s)$, $s' \sim P(\cdot | s, a)$, TD updates:

$$\hat{V}^\pi(s) \leftarrow \hat{V}^\pi(s) + \eta \left(r + \gamma \hat{V}^\pi(s') - \hat{V}^\pi(s) \right)$$

On-policy: data is generated from the policy π itself

Off-policy: data is generated from policy π_b where $\pi_b \neq \pi$

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Q: is Q-learning off-policy or on-policy?

Motivation for off-policy evaluation

Counterfactual: what would happen if I did something different?

Off-policy TD Learning

Setting: data is generated by π_b , but we want to estimate V^π for some $\pi \neq \pi_b$

Key trick: importance weighting

$$V^\pi(s) = \mathbb{E}_{a \sim \pi(\cdot|s)} \left(r(s, a) + \gamma \mathbb{E}_{s' \sim P(\cdot|s, a)} V^\pi(s') \right)$$

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Off-policy TD Learning

Setting: data is generated by π_b , but we want to estimate V^π for some $\pi \neq \pi_b$

Key trick: importance weighting

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Importance weight

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Now action is
sampled from π_b

Importance weight

Off-policy TD Learning

Given (s, a, r, s') , where $a \sim \pi_b(\cdot | s)$, $s' \sim P(\cdot | s, a)$,

Off-policy TD updates as follows:

$$\hat{V}^\pi(s) \Leftarrow \hat{V}^\pi(s) + \eta \frac{\pi(a | s)}{\pi_b(a | s)} \left(r + \gamma \hat{V}^\pi(s') - V^\pi(s) \right)$$

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Case 1: $\pi(a | s)$ is large but $\pi_b(a | s)$ is small

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Case 1: $\pi(a | s)$ is large but $\pi_b(a | s)$ is small

Case 2: $\pi(a | s)$ is small but $\pi_b(a | s)$ is large

Summary

Q-Learning: online algorithm that learns Q^* (bootstrapping)

Exploration & Exploitation tradeoff: ϵ -greedy is an effective heuristic

Off-policy policy evaluation: importance weighting
(also known as inverse probability weighting in causal inference)

Feedback and Attendance

<https://bit.ly/4gYn2nC>

